Distance Vector Routing: RIP & EIGRP

RIP – Routing Information Protocol
V1 – largely absent from modern networks
V2 – deprecated, but still in some use

RIPv1 Limitations
Sends a full routing table update at a fixed interval (every 30 seconds) (whether changes were made or not)
Does not understand subnet masks

Split Horizon - a route cannot be advertised via an interface if that same interface is the one that learned about the route in the first place

Route Poisoning – to notify the network explicitly that it is down by sending RIP route information set to metric 16. By route poisoning, network down can be detected more quickly than timer-based detection of network down, and unnecessary route information can be removed from the routing table.

when routers agree on the current state of the network, they have reached a state of convergence. A major reason you don't see a lot of DV routing out there is that DV protocols are slow to converge (even in a lab environment).

The metric with RIP is Hop Count - There is a maximum hop count of 15, a poisoned route will advertise itself as being 16 hops away to let the other routers know this is an invalid route

Hop count is a poor metric, as it does not take link speed into account
A 56k link and a 1 gbps link would be seen as the same – 1 hop away

Sending the frame over RIP will be inefficient if the connections have multiple speeds as the rip protocol makes no differentiation other than Hop Count (such as speed or AD)